CPSC 304 Project Cover Page

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above. (In the case of Project Milestone 0, the main purpose of this page is for you to let us know your e-mail address, and then let us assign you to a TA for your project supervisor.)
In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

1. Repository Link

Link to our team's repo:
https://github.students.cs.ubc.ca/CPSC304-2025W-T1/team_30/tree/main

2. SQL File Assertions

We only require assertions for global constraints that cannot be expressed using row-level constraints like primary/foreign keys, NOT NULL, UNIQUE, or CHECK. In our schema, constraints such as "checkpoint must have at least one objective" and "each save file has exactly one current checkpoint" fall into this category, whereas conditions like price > 0 and qty > 0 are enforced with simple CHECK constraints and do not require assertions. For the full SQL file, please check the project repo.

3. Project Description

This application is a web-based management system for an RPG video game database. It allows game administrators to manage player accounts, track save files, monitor achievement progress, and maintain in-game world data, including characters, items, locations, and objectives.

Key features include user management, save file creation and analysis, character search for PCs and NPCs, achievement tracking, item inventory management with statistics, and aggregated user reports. The system solves the problem of centralising RPG backend data management, enabling progress syncing across sessions and potential integration with live services like PlayStation accounts.

Schema Differences: Previous vs Final

1. Table Renamed: Boost to ItemBoost for better clarity and naming consistency with ItemSaleValue.
2. Data Type Fix: VARCHAR to VARCHAR2 throughout (Oracle SQL requirement).
3. ON DELETE Actions Added:
    a. ON DELETE CASCADE on most foreign keys to automatically remove dependent records

    b. ON DELETE SET NULL on Objectives' FKs to Locations/Items (preserves objectives if location/item deleted)

4. Why these changes?

Oracle compatibility required VARCHAR2 and quoting reserved words. Renaming Boost improved schema readability. ON DELETE actions ensure proper referential integrity handling. Without them, deleting parent records would fail or leave dangling data.

4. Relational Schema

1. **CharTypes** (chr_type: char[20], base_hp: int, base_damage: int)
    a. Primary key: chr_type
    b. Not null: base_hp, base_damage

2. **ItemSaleValue** (item_rarity: char[20], sale_value: int)
    a. Primary key: item_rarity
    b. Not null: sale_value

3. **ItemBoost** (item_rarity: char[20], item_type: char[20], boost: int)
    a. Primary key: (item_rarity, item_type)
    b. Not null: boost

4. **Difficulty** (difficulty: char[20], hp_mult: int, enemy_dmg_mult: int)
    a. Primary key: difficulty
    b. Not null: hp_mult, enemy_dmg_mult

5. **Users** (username: char[20], first_name: char[20], created_on: date, email: varchar2[255])
    a. Primary key: username
    b. Not null: email
    c. Unique: email

6. **Locations** (location_name: char[20], story: varchar2[4000], biome: char[10], landmark: char[20])
    a. Primary key: location_name

   b. Not null: biome, landmark
   c. Unique: (biome, landmark)

7. **Items** (item_name: char[20], item_rarity: char[20], item_type: char[20])
   a. Primary key: item_name
   b. Not null: item_rarity, item_type
   c. Foreign key: item_rarity -> ItemSaleValue (ON DELETE CASCADE), (item_rarity, item_type) -> ItemBoost (ON DELETE CASCADE)

8. **Objectives** (obj_name: char[20], description: varchar2[4000], location_name: char[20], item_name: char[20])
   a. Primary key: obj_name
   b. Foreign key: location_name -> Locations (ON DELETE SET NULL), item_name -> Items (ON DELETE SET NULL)

9. **NPCs_IsAt** (chr_name: char[20], chr_type: char[20], bg_story: char[20], location_name: char[20])
   a. Primary key: chr_name
   b. Not null: chr_type
   c. Foreign key: location_name -> Locations (ON DELETE SET NULL), chr_type -> CharTypes (ON DELETE CASCADE)

10. **PCs** (chr_name: char[20], chr_type: char[20], bg_story: char[20])
    a. Primary key: chr_name
    b. Not null: chr_type
    c. Foreign key: chr_type -> CharTypes (ON DELETE CASCADE)

11. **Checkpoints_IsAt** (checkpt_number: int, description: varchar2[4000], location_name: char[20])
    a. Primary key: checkpt_number
    b. Not null: location_name
    c. Foreign key: location_name -> Locations (ON DELETE CASCADE)

12. **SaveFiles_IsAt** (username: char[20], created_on: date, difficulty: char[20])

a. Primary key: (username, created_on)
b. Not null: difficulty
c. Foreign key: username -> Users (ON DELETE CASCADE), difficulty -> Difficulty (ON DELETE CASCADE)

13. **Achievements_monitors** (achievement_name: char[20], description: varchar2[4000], username: char[20], obj_name: char[20])
    a. Primary key: achievement_name
    b. Not null: description, username, obj_name
    c. Foreign key: username -> Users (ON DELETE CASCADE), obj_name -> Objectives (ON DELETE CASCADE)

14. **Completes** (username: char[20], obj_name: char[20])
    a. Primary key: (username, obj_name)
    b. Not null: username, obj_name
    c. Foreign key: username -> Users (ON DELETE CASCADE), obj_name -> Objectives (ON DELETE CASCADE)

15. **Earned** (achievement_name: char[20], username: char[20], date_earned: date)
    a. Primary key: (achievement_name, username)
    b. Not null: achievement_name, username, date_earned
    c. Foreign key: achievement_name -> Achievements_monitors (ON DELETE CASCADE), username -> Users (ON DELETE CASCADE)

16. **"Current"** (username: char[20], created_on: date, chr_name: char[20])
    a. Primary key: (username, created_on, chr_name)
    b. Foreign key: (username, created_on) -> SaveFiles_IsAt (ON DELETE CASCADE), chr_name -> PCs (ON DELETE CASCADE)

17. **Contains** (username: char[20], created_on: date, item_name: char[20], quantity: int)
    a. Primary key: (username, created_on, item_name)

b. Foreign key: (username, created_on) -> SaveFiles_IsAt (ON DELETE CASCADE), item_name -> Items (ON DELETE CASCADE)

18. **HasAPrerequisite** (checkpt_number: int, obj_name: char[20])
    a. Primary key: (checkpt_number, obj_name)
    b. Foreign key: checkpt_number -> Checkpoints_IsAt (ON DELETE CASCADE), obj_name -> Objectives (ON DELETE CASCADE)

5. Screenshots of Seed Data

- CharTypes:

chr_type	base_hp	base_damage
Warrior	100	25
Mage	70	30
Rogue	80	20
Healer	90	15
Tank	120	18

- ItemSaleValue:

item_rarity	sale_value
Common	10
Uncommon	25
Rare	50
Epic	100
Legendary	250

- ItemBoost:

item_rarity	item_type	boost
Common	Weapon	5
Common	Armor	3
Uncommor	Weapon	10
Uncommor	Armor	12
Rare	Weapon	18
Rare	Armor	15
Epic	Weapon	25
Epic	Armor	30
Legendary	Weapon	45
Legendary	Armor	40

- Difficulty:

difficulty	hp_mult	enemy_dmg_mult
Easy	1	1
Normal	2	2
Hard	3	3
Expert	4	5
Nightmare	5	7

- Users:

username	first_name	created_on	email
player1	John	2024-01-15	john@email.com
player2	Sarah	2024-02-20	sarah@email.com
player3	Mike	2024-03-10	mike@email.com
player4	Emma	2024-04-05	emma@email.com
player5	Alex	2024-05-12	alex@email.com

- Locations:

location_name	story	biome	landmark
Dark Forest	A mysterious forest shrouded in eternal twilight, home to ancient secrets.	Forest	Ancient Oak
Crystal Cave	Glowing crystals illuminate this underground wonder, pulsing with magic.	Cave	Crystal Shrine
Desert Ruins	Ancient civilization remnants half-buried in endless golden sands.	Desert	Pyramid
Mountain Peak	The highest point in the realm, where storms and legends are born.	Mountain	Summit Temple
Coastal Village	A peaceful fishing village where adventurers begin their journey.	Coast	Lighthouse

- Items:

item_name	item_rarity	item_type
Iron Sword	Common	Weapon
Steel Armor	Uncommon	Armor
Magic Staff	Rare	Weapon
Dragon Shield	Epic	Armor
Excalibur	Legendary	Weapon

- Objectives:

obj_name	description	location_name	item_name
Find Key	Search the forest floor for the hidden key.	Dark Forest	Iron Sword
Defeat Boss	Defeat the Crystal Guardian in combat.	Crystal Cave	Magic Staff
Solve Puzzle	Decode the ancient hieroglyphics.	Desert Ruins	
Rescue NPC	Save the stranded climber from the cliff.	Mountain Peak	
Collect Gems	Gather 10 sea gems from the shore.	Coastal Village	Dragon Shield
Find Artifact	Locate the ancient artifact in the woods.	Dark Forest	
Map Cave	Create a complete map of the cave system.	Crystal Cave	

- NPCs_IsAt:

chr_name	chr_type	bg_story	location_name
Village Elder	Healer	Wise	Coastal Village
Guard Captain	Warrior	Loyal	Dark Forest
Merchant Tom	Rogue	Shady	Desert Ruins
Wizard Merlin	Mage	Ancient	Crystal Cave
Mountain Guide	Tank	Sturdy	Mountain Peak

- PCs:

chr_name	chr_type	bg_story
Hero1	Warrior	Noble
Hero2	Mage	Scholar
Hero3	Rogue	Thief
Hero4	Healer	Priest
Hero5	Tank	Dweller

- Checkpoints_IsAt:

checkpt_number	description	location_name
1	The journey begins at the village.	Coastal Village
2	Deep within the dark woods.	Dark Forest
3	The crystal chamber entrance.	Crystal Cave
4	The desert oasis checkpoint.	Desert Ruins
5	The final ascent begins.	Mountain Peak

- SaveFiles_IsAt:

username	created_on	difficulty
player1	2024-01-15	Normal
player1	2024-06-01	Hard
player2	2024-02-20	Hard
player2	2024-07-15	Expert
player2	2024-08-20	Nightmare
player3	2024-03-10	Easy
player4	2024-04-05	Expert
player5	2024-05-12	Nightmare

- Achievements_monitors:

achievement_name	description	username	obj_name
First Steps	Complete your first objective.	player1	Find Key
Cave Explorer	Defeat the Crystal Guardian.	player2	Defeat Boss
Puzzle Master	Solve an ancient puzzle.	player3	Solve Puzzle
Hero	Rescue someone in need.	player4	Rescue NPC
Collector	Gather all the gems.	player5	Collect Gems

- Completes:

username	obj_name
player1	Find Key
player1	Find Artifact
player2	Defeat Boss
player2	Map Cave
player3	Solve Puzzle
player4	Rescue NPC
player5	Collect Gems

- Earned:

achievement_name	username	date_earned
First Steps	player1	2024-01-20
Cave Explorer	player2	2024-02-25
Puzzle Master	player3	2024-03-15
Hero	player4	2024-04-10
Collector	player5	2024-05-20

- Current:

username	created_on	chr_name
player1	2024-01-15	Hero1
player1	2024-06-01	Hero2
player2	2024-02-20	Hero2
player3	2024-03-10	Hero3
player4	2024-04-05	Hero4
player5	2024-05-12	Hero5

- Contains:

username	created_on	item_name	quantity
player1	2024-01-15	Iron Sword	1
player1	2024-01-15	Steel Armor	2
player1	2024-06-01	Magic Staff	1
player2	2024-02-20	Steel Armor	1
player2	2024-02-20	Magic Staff	1
player2	2024-02-20	Dragon Shield	1
player3	2024-03-10	Magic Staff	1
player4	2024-04-05	Dragon Shield	2
player5	2024-05-12	Excalibur	1
player5	2024-05-12	Dragon Shield	1

- HasAPrerequisite:

checkpt_number	obj_name
1	Find Key
2	Find Key
3	Defeat Boss
4	Solve Puzzle
5	Rescue NPC

6. Required Queries

Query #	Query Type	File	Lines	SQL Query
1	INSERT	ketan-routes.js	19	INSERT INTO SaveFiles_IsAt (username, created_on, difficulty) VALUES (:username, SYSDATE, :difficulty)
2	DELETE	samarth-routes.js	31	DELETE FROM Users WHERE TRIM(username) = :username
3	UPDATE	samarth-routes.js	91	UPDATE Items SET item_rarity = :item_rarity, item_type = :item_type WHERE TRIM(item_name) = :item_name

4	SELECTIO N	ketan-routes.js	148	Dynamic WHERE with AND/OR: SELECT ... FROM NPCs_IsAt WHERE 1=1 AND (condition1 AND/OR condition2 ...)
5	PROJECTI ON	ketan-routes.js	284	SELECT ${selectColumns} FROM Users ORDER BY TRIM(username) (dynamic column selection)
6	JOIN	samarth-routes.js	166	SELECT ... FROM Earned e JOIN Achievements_monitors am ON ... JOIN Users u ON ... JOIN Objectives o ON ...
7	GROUP BY	ketan-routes.js	347	SELECT TRIM(i.item_rarity), COUNT(*), ... FROM Items i JOIN ItemSaleValue isv ON ... GROUP BY TRIM(i.item_rarity), isv.sale_value
8	HAVING	samarth-routes.js	240	SELECT ... FROM Users u LEFT JOIN SaveFiles_IsAt sf ON ... GROUP BY ... HAVING COUNT(sf.created_on) >= :threshold

9	NESTED AGGREGATION	ketan-routes.js	378	SELECT ... FROM Difficulty d LEFT JOIN SaveFiles_IsAt sf ON ... LEFT JOIN (SELECT username, created_on, COUNT(*) AS item_count FROM Contains GROUP BY username, created_on) item_counts ON ... GROUP BY ...
10	DIVISION	samarth-routes.js	279	SELECT ... FROM Completes c JOIN Users u ON ... JOIN Objectives o ON ... WHERE TRIM(o.location_name) = :location GROUP BY ... HAVING COUNT(DISTINCT TRIM(c.obj_name)) = :objectiveCount